

Time-Series Forecasting of Appliance Energy Use

Benchmarks, SARIMA/SARIMAX, feature-based machine learning and Chronos-Bolt

1. Introduction

Accurate short-term household forecasts of energy consumption enable demand response, appliance scheduling and smart-home control. The problem is that appliance consumption features both predictable recurring patterns and unpredictable switches and spikes. In this work, we ask a practical question of how much forecasting power is gained from simple seasonal rules by moving to statistical, feature-based and foundation approaches, and is the gain worth the complexity?

We evaluate the accuracy of forecasts of hourly average appliance energy consumption (in Wh) 24 hours ahead. We compare five benchmarks against the seasonal ARIMA/SARIMAX, Histogram Gradient Boosting and Random Forests and the zero-shot Chronos-Bolt model. Accuracy is measured on a rolling-origin holdout using MAE, RMSE, MASE, sMAPE and bias, alongside the coverage of intervals if available. The design follows the forecasting literature recommendation to use a holdout rather than random splits of data, so that the test set represents future observations relative to the training set (Hyndman and Athanasopoulos, 2021).

2. Dataset and preprocessing

The UCI Appliances Energy Prediction dataset contains 19,735 ten-minute observations made between 11 January and 27 May 2016 (Candanedo, 2017; Candanedo, Feldheim and Deramaix, 2017). Appliance and lighting energy, nine pairs of indoor temperature/humidity sensors, and outdoor weather measurements are available for consideration. Timestamp auditing revealed that there were no duplicates, no missing timestamps, and no missing values. Six ten-minute observations were combined to form hourly averages, one partially empty terminal hour was excluded, yielding 3,289 observations.

The hourly appliance demand is highly skewed with a mean of 97.68 Wh, a median of 63.33, a standard deviation of 81.02, a minimum of 28.33, a maximum of 608.33 and a skewness of 2.389. The discrepancy between the mean and the median and the presence of extreme outliers is important in this case because of the use of squared-error metrics penalizing peak estimation errors.

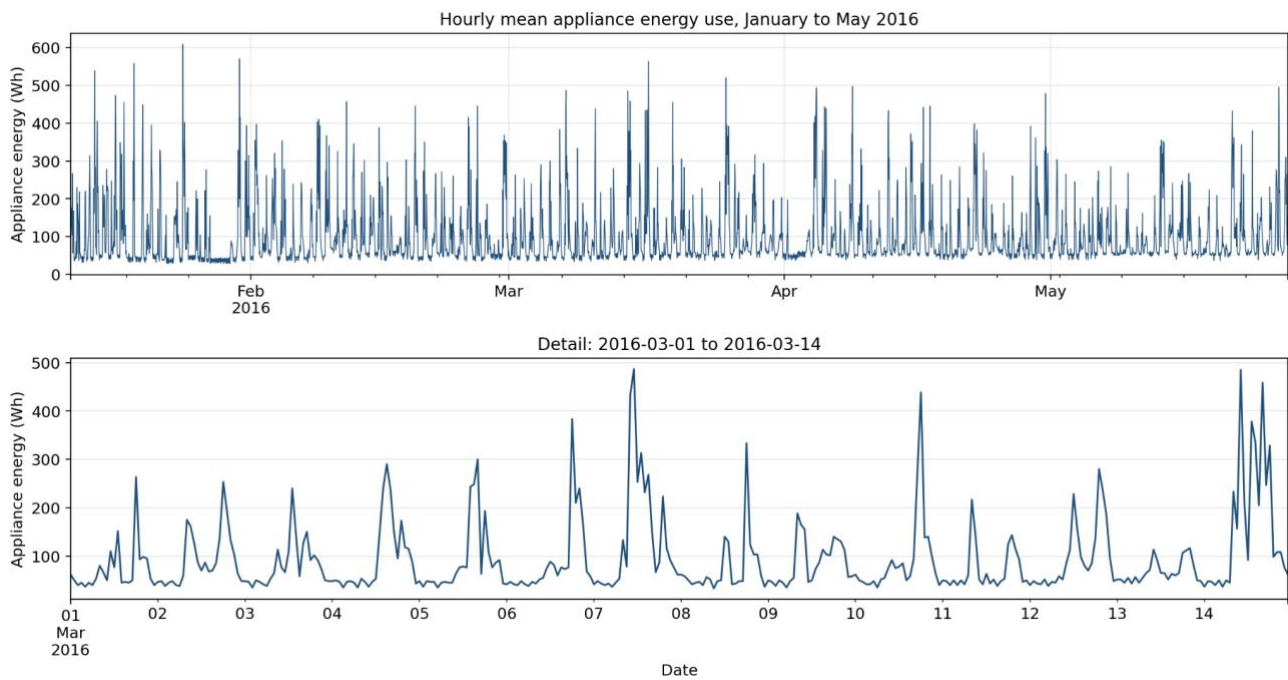


Figure 1. Hourly appliance-energy demand from January-May 2016 and a two-week detail window. Repeated low-demand periods are punctuated by irregular high-demand events.

3. Exploratory analysis, seasonality and stationarity

The series shows clear intraday structure. Mean demand is lowest overnight, rises through the morning and reaches its largest average peak at approximately 18:00. Weekend profiles differ from weekdays, particularly around late morning, indicating behavioural as well as purely clock-time effects. The decomposition estimated a 24-hour seasonal strength of 0.474 and a 168-hour weekly strength of 0.298, whereas trend strength was only 0.046. Thus seasonality is meaningful but does not dominate the process; a large fraction of the variance is irregular switching behaviour.

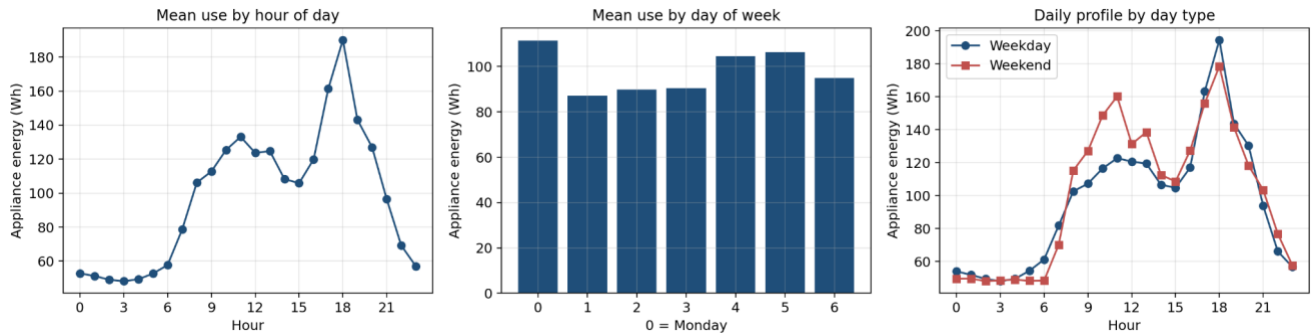


Figure 2. Mean appliance demand by hour, day of week and weekday/weekend status. The strongest recurring pattern is the daily cycle, with additional weekly behavioural variation.

Autocorrelation confirms the visual evidence. ACF values were 0.581 at lag 1, 0.304 at lag 24, 0.221 at lag 48 and 0.306 at lag 168. Repeated peaks every 24 hours motivate daily seasonal models, while the weekly correlation motivates a 168-hour seasonal-naïve benchmark. PACF is strongest at the first few lags and shows smaller seasonal structure, consistent with a combination of short-memory autoregression and periodic dependence.

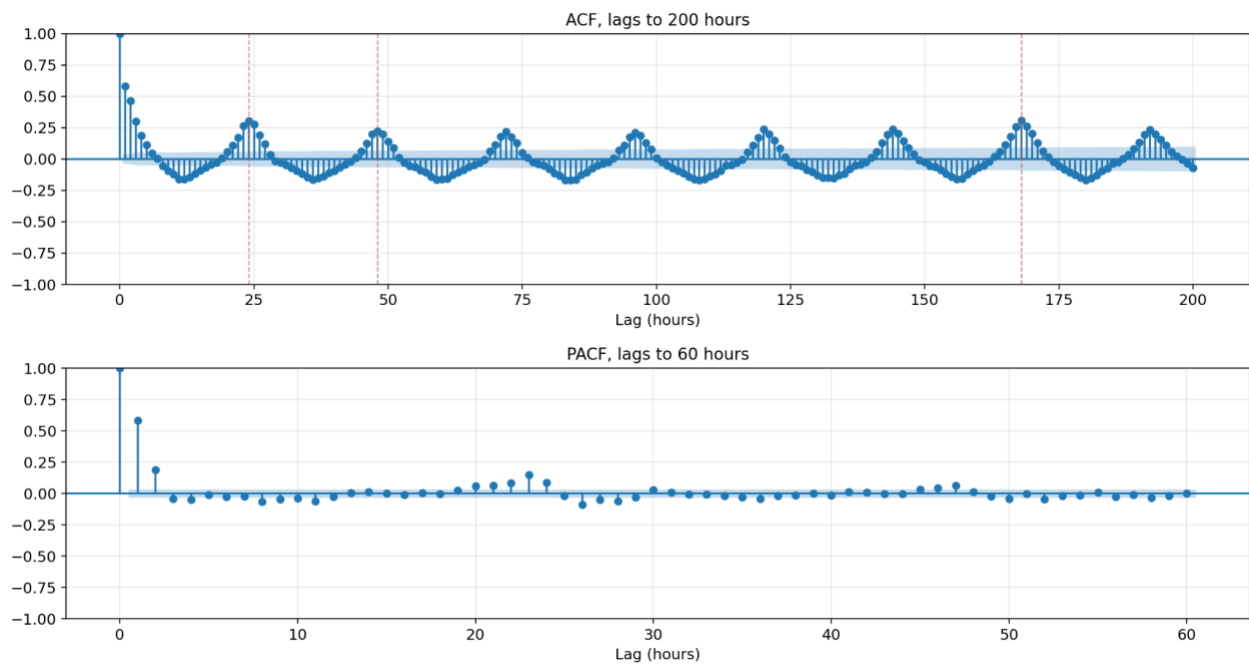


Figure 3. ACF to 200 hours and PACF to 60 hours. Repeated ACF peaks at daily multiples and at one week provide direct evidence of seasonal dependence.

3.1 Stationarity

The level series was tested using complementary ADF and KPSS tests. ADF = -9.085 ($p = 3.97 \times 10^{-15}$) rejects a unit root, while KPSS = 0.041 ($p = 0.10$) does not reject stationarity. These results support level stationarity, but stationarity does not imply an absence of seasonality: a stationary process may retain predictable periodic autocorrelation. Consequently, model selection was allowed to determine whether differencing and seasonal differencing improved fit rather than imposing them solely from the hypothesis tests.

4. Forecasting design and benchmark models

The final 14 days (336 hours) were held out. At the first origin, 2,953 hourly observations were available for training. The evaluation then used 14 non-overlapping rolling origins, each producing a 24-hour forecast. This design preserves temporal order and evaluates the exact horizon requested by the assignment. The MASE scaling denominator, calculated from training data only, was 53.359 Wh.

4.1 Benchmark definitions and results

The mean forecast is a historical training mean, the naive repeats the last observation, while the daily and weekly seasonal naive counterparts repeat the same hour of the day and 24 and 168 hours ago, and finally, the drift forecasts a linear trend from historical changes. These methods are all simplistic enough that the additional complexity of the other proposed process should prove beneficial (Hyndman and Athanasopoulos, 2021).

The weekly seasonal naive forecast was the best-performing benchmark, with an MAE of 42.92Wh, which is approximately 11.3% better than the daily seasonal naive. The difference highlights the value of the same-hour next-week observation as a predictor above and beyond the same hour of the day. However, the weekly versus daily seasonal naive Diebold-Mariano test was not significant at $p = 0.496$, and the weekly seasonal naive had the MAE between 7.01 and 97.64Wh, with a standard deviation of 28.24Wh depending on the forecast origin. Thus, while the weekly seasonal naive was overall the best, the difference was not significant enough to conclude that it had a different underlying process than the daily seasonal naive.

The naive and drift forecasts suffered from severe positive biases of approximately 115Wh, which was explained by the time of day when the forecasts' origins lay. Namely, the 17:00 naive observation was right before the 18:00 electric kettle peak, and thus, it carried a significant 215Wh on average for the forecasted value compared to the 100Wh on average for the held-out series.

This is why the forecast underperforms, as the forecast origin lies at a local maximum for the kettle use. Nevertheless, the naive forecast highlights the overall lack of long-term dependency in the electric kettle use, with the naive (daily) forecasting displaying a strong mean reversion at best and random walk at worst.

5. SARIMA and SARIMAX

The required non-seasonal search evaluated all $p = 0 \dots 6$, $d = 0 \dots 2$ and $q = 0 \dots 6$ combinations (147 candidate orders) by AIC. The executed seasonal stage carried the three strongest base orders forward and combined each with all (P,D,Q) in $\{0,1\}^3$ at seasonal period $s = 24$, adding 24 seasonal fits. The selected model was SARIMA(2,0,6)(0,1,1)₂₄ with AIC 32,137.2. Its approximately 807-point AIC improvement over the best non-seasonal fit is strong evidence that explicit daily seasonality materially improves the likelihood-based model.

Residual diagnostics distinguish model fit from forecast accuracy. Ljung-Box p-values of 0.675, 0.406 and 0.142 at lags 24, 48 and 168 do not reject residual independence, and residual ACF at lag 24 is only +0.023. Thus most linear seasonal dependence is removed. Nevertheless, Jarque-Bera = 10,046.3 ($p \approx 0$) and the Q-Q plot show severe non-normality and heavy positive tails. The model therefore captures autocorrelation better than it captures the distribution of rare demand spikes; Gaussian prediction intervals should be interpreted with this limitation in mind.

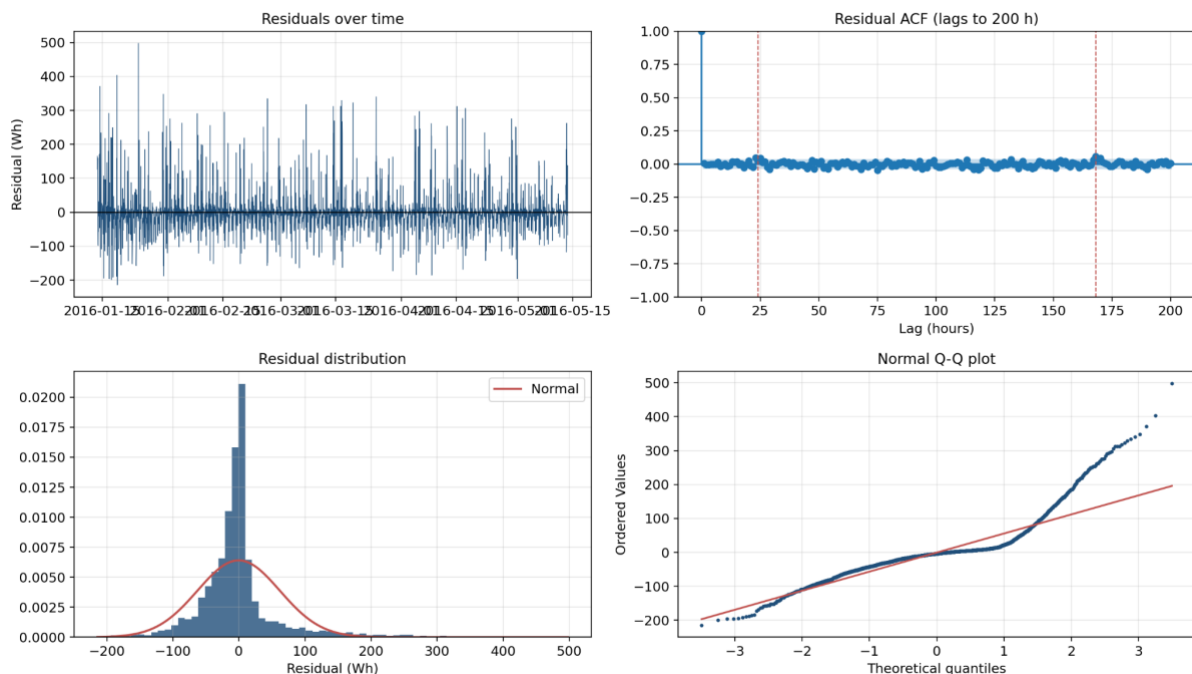


Figure 4. SARIMA residual diagnostics. Residual autocorrelation is small, but the distribution and Q-Q plot reveal strong non-normality and heavy tails.

Plain SARIMA has the best MAE and MASE, improving MAE by 13.7% over weekly seasonal naive. Calendar SARIMAX has a slightly lower RMSE (63.53 Wh), suggesting fewer or smaller large errors, but worse typical absolute error. Neither weather nor calendar covariates improve MAE, so their added complexity is not justified by point accuracy. The 95% intervals are well calibrated overall (PICP ≈ 0.95), although they are wide (about 273–278 Wh). SARIMA versus weekly seasonal naive is not statistically significant (DM $p = 0.340$).

6. Covariates, feature engineering and machine learning

Feature engineering combined deterministic calendar variables, appliance lags, rolling and same-hour historical statistics, lagged indoor temperature/humidity variables, lights and weather. A critical design distinction is forecast availability: calendar variables and historical lags are known at origin, whereas realised future sensor and weather values are not. Future observed covariates from the test set therefore create a conditional rather than an operational forecast.

6.1 Which feature groups add information?

The ablation study shows that temporal structure is more valuable than raw environmental data. Calendar alone achieved MAE 40.91 Wh, target rolling features 41.25 Wh and target lags 41.75 Wh. Indoor sensors, lights and weather were progressively weaker when used alone (44.21, 46.18 and 52.62 Wh). When groups were added cumulatively, calendar plus rolling target features produced the best compact subset at 37.87 Wh; adding target lags changed little (37.89 Wh), while adding indoor sensors and lights degraded performance. This is evidence of redundancy/noise rather than a simple “more features is better” effect.

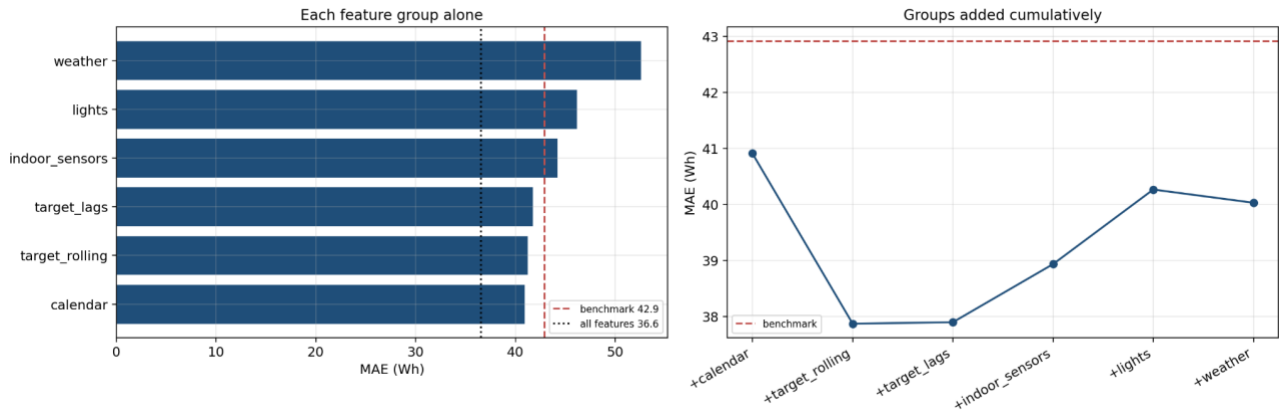


Figure 5. Feature-group ablation. Temporal/calendar information contributes most useful signal; additional sensor, lighting and weather groups do not monotonically improve accuracy.

Mutual-information and permutation analyses support the same interpretation. The most informative variables include hour of day, seven-day same-hour mean and standard deviation, lag 24, lag 168 and cyclical hour features. This aligns with the strong daily/weekly structure observed in the ACF and explains why feature engineering adds value primarily by representing recurring behaviour rather than by adding contemporaneous weather.

6.2 Machine-learning forecasts

*Regime B uses future covariate information that is not guaranteed to be known at the forecast origin; its 36.55 Wh MAE should therefore be interpreted as conditional performance. Random Forest is an ensemble of decorrelated regression trees (Breiman, 2001), while Histogram Gradient Boosting sequentially corrects residual structure using nonlinear decision trees.

7. Foundation model: Chronos-Bolt

Chronos-Bolt was evaluated zero-shot: no parameters were trained on this household. At each rolling origin the model received a 512-hour historical context and produced a 24-hour probabilistic forecast. Chronos is a pretrained probabilistic time-series framework designed to transfer forecasting structure across datasets, making zero-shot evaluation a meaningful contrast with locally fitted models (Ansari et al., 2024). Context sensitivity in the notebook showed 512 hours to be slightly better than 168, 336 and 1024 hours by MAE. Chronos achieved MAE 33.58 Wh, RMSE 67.53 Wh, MASE 0.629, sMAPE 26.01% and bias -18.05 Wh. It therefore has the lowest MAE and MASE of the tested methods, improving MAE by 21.8% relative to weekly seasonal naive and by about 8.1% relative to conditional HGB Regime B. Its 80% interval achieved PICP 0.815 with mean width 94.36 Wh, close to nominal coverage.

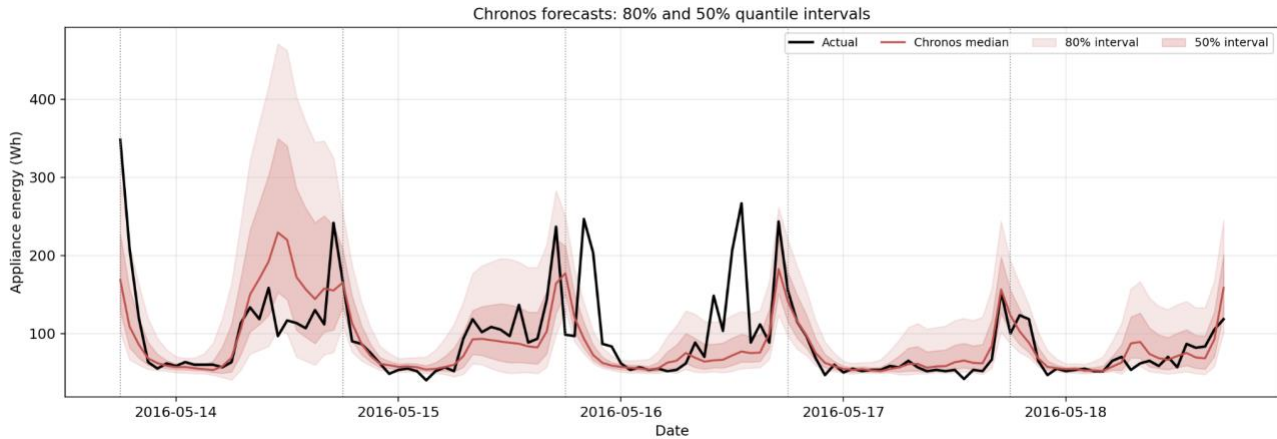


Figure 4. Chronos-Bolt median forecasts with 50% and 80% quantile intervals. Uncertainty expands where spikes are plausible, although some realised peaks remain poorly captured.

7.1 Accuracy is not the whole story

The order of metrics is not the same. For example, the lowest RMSE (61,51 Wh) is found in HGB Regime B, while Chronos has the lowest MAE. This suggests that Chronos was better calibrated for typical and conservative hours but performed significantly worse on some extreme hours. Indeed, for the 37 test hours with more than 200 Wh of mean demand, the actual mean demand was 278,0 Wh, while Chronos only predicted 112,8 Wh on average and had a peak-hour MAE of 166,1 Wh. HGB Regime B predicted 153,2 Wh with a peak-hour MAE of 125,3 Wh. Calibration slopes confirm this version of the underestimation of extreme values, being equal to 0.23 for Chronos and 0.35 for Regime B, while the best would be 1, and they are significantly lower (0.30 and 0.31) for Random Forest and standard HGB.

Calibration: a slope below 1 means peaks are under-predicted

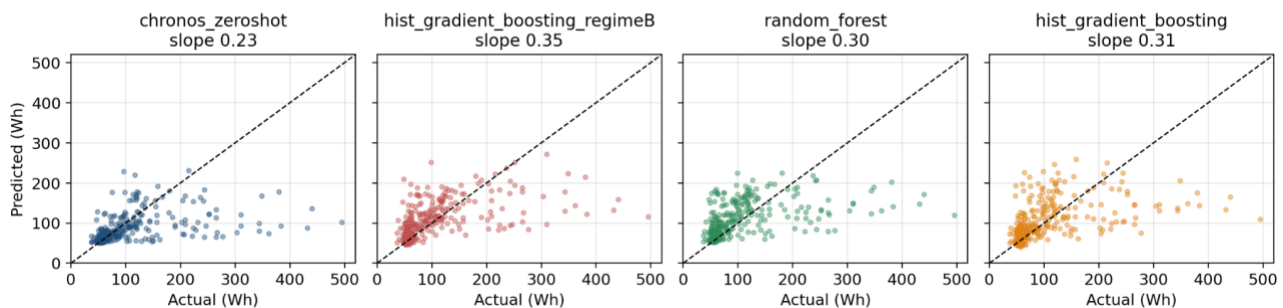


Figure 7. Predicted versus actual demand for advanced models. Slopes far below one show systematic under-representation of high-demand peaks.

The top ranking is also statistically uncertain. Chronos versus weekly seasonal naive gives DM $p = 0.0722$ and Chronos versus HGB Regime B $p = 0.2098$. Origin-level Chronos MAE has SD 22.46 Wh, much larger than the 2.97 Wh gap between the top two models. The defensible conclusion is therefore that Chronos has the best observed MAE, not that it is decisively superior on this 14-day holdout.

8. Overall evaluation and required analytical questions

The compact calendar plus rolling-target feature set achieved an MAE of 37.87 Wh using only 33 features, which would rank third overall and outperform the full 76-feature model (MAE 40.03 Wh). This suggests that the most useful predictive information comes from time-of-day structure and the smoothed recent demand level, while the additional sensor and weather channels mainly introduce redundancy and variance without enough new signal to improve generalisation.

Model	Family	MAE	RMSE	MASE	sMAPE	Bias
Chronos zero-shot	Foundation	33.58	67.53	0.629	26.01	-18.05
HGB Regime B*	ML/conditional	36.55	61.51	0.685	30.55	3.00
SARIMA	Statistical	37.06	64.08	0.695	30.18	-3.15
SARIMAX weather	Statistical	38.33	64.24	0.718	32.77	-3.32
Random Forest	ML	39.37	64.06	0.738	31.70	2.22
SARIMAX calendar	Statistical	39.66	63.53	0.743	33.62	3.07
Histogram Gradient Boosting	ML	40.03	65.80	0.750	32.53	0.73
Weekly seasonal naive	Benchmark	42.92	79.67	0.804	32.67	-12.62
Daily seasonal naive	Benchmark	48.39	85.79	0.907	35.57	1.67
Mean	Benchmark	50.01	73.99	0.937	46.24	-3.06
Naive	Benchmark	143.38	170.62	2.687	87.07	114.67
Drift	Benchmark	143.94	171.26	2.698	87.23	115.31

Table 1. Common pooled metrics over the same 336 held-out hours. Lower is better for MAE, RMSE, MASE and sMAPE; bias should be close to zero. *Regime B is conditional because future covariate availability is not guaranteed.

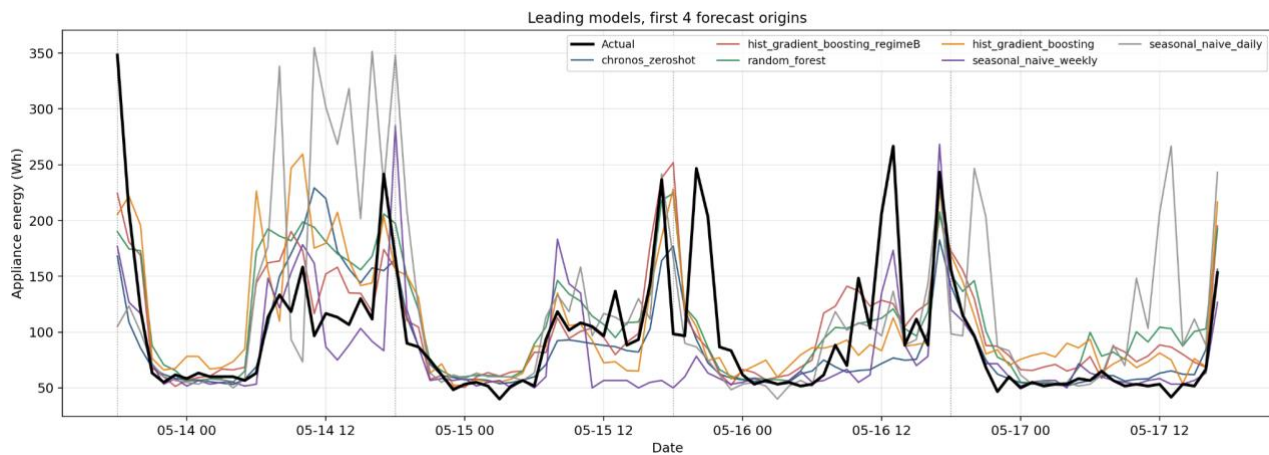


Figure 5. Leading benchmark, machine-learning and foundation-model forecasts over four origins. The common failure mode is visible: broad daily structure is captured, while abrupt peaks are smoothed.

Q1. Strongest benchmark. Weekly seasonal naive is strongest (MAE 42.92 Wh). This shows that appliance use contains repeatable weekly routines in addition to daily seasonality. Its advantage over daily seasonal naive is directionally meaningful but not statistically significant on 14 origins.

Q2. Does SARIMAX improve on the benchmark? Yes for the seasonal ARIMA family: plain SARIMA reduces MAE to 37.06 Wh and removes most residual seasonal autocorrelation. However, weather/calendar exogenous variables do not improve MAE. Daily seasonality and linear dependence are captured better than rare peak behaviour.

Q3. Do engineered features improve ML? Yes, but selectively. Rolling target history, lags and time-of-day features are most useful. Calendar plus rolling history is the best compact subset; adding all sensor/weather variables worsens accuracy, showing diminishing returns and redundancy.

Q4. Does the foundation model justify its complexity? Chronos has the lowest observed MAE (33.58 Wh) and is 21.8% better than the strongest benchmark, but it is not significantly better than weekly seasonal naive at the 5% level and is not significantly better than Regime B. Its advantage is therefore promising rather than decisive, particularly given its heavier dependency and weaker peak calibration.

Q5. Which covariates are genuinely available? Calendar variables, historical appliance values, lags and rolling statistics are available at forecast origin. Realised future indoor temperature, humidity, lighting and weather from the test set are not. Supplying those values produces a conditional forecast unless they themselves come from separate operational forecasts. This is why Regime B should not be compared uncritically with fully deployable models.

The Regime A feature set was explicitly tested for temporal leakage and passed all eight checks. In the decisive test, a feature row rebuilt using history truncated at its own timestamp was identical to the row generated when the full dataset was available, confirming that future observations did not enter the legitimate predictors. As a positive control, a deliberately leaked feature correlated 0.585 with the target, compared with 0.435 for the strongest legitimate feature, demonstrating that the leakage checks had meaningful detection power.

Q6. Which model should be recommended? For a practical smart-home deployment, SARIMA is the preferred compromise. It materially improves on the seasonal benchmark, is interpretable, uses only historical demand, has approximately nominal 95% uncertainty coverage and is substantially lighter to maintain than a foundation model. Chronos is the accuracy-oriented alternative when the infrastructure cost is acceptable. Standard deployable ML remains useful, but its point accuracy is weaker than SARIMA in this experiment; the stronger Regime B score depends on conditional future covariates.

9. Limitations and future improvements

The most significant limitation of this work is the restricted evaluation set. The data represent one house over the course of approximately 4.5 months, and the holdout set only contains 14 days of origins, so extreme days have a disproportionate impact on the ranking. A rolling-origin evaluation across different seasons and more households would be required to reliably demonstrate generalisation. Secondly, it is crucial to remember that all good probabilistic forecasts will contract the variance around the mean, thus, for the next version of the work, it is essential to explore alternative peak-aware losses, two-stage models that learn the probability of a peak occurrence alongside its magnitude, occupancy/activity-aware features, and/or model assembling that combines seasonal-statistical models with nonlinear components.

Additionally, the treatment of covariates in the current work is far too simplistic. It is entirely possible to replace the weather covariates with a forecast, but temperature and humidity inside may benefit from separate predictive models. Nested time-series cross-validation would be required to reduce optimism in the hyperparameter selection, and calibration using either conformal or quantile forecasts would be a more robust choice than assuming normality of residuals, which are clearly not normally distributed. Finally, the presence of serial correlation in the residuals of the advanced models suggests that hybrid or ensemble models may still benefit from additional components that exploit structure at multiple levels.

10. Conclusion

Appliance demand is stationary in level but characterized by highly structured daily and weekly patterns with significant irregular switching noise. In this regard, the weekly seasonal naive is a challenging benchmark, not an easy competitor. SARIMA (2,0,6)(0,1,1)₂₄ achieves the best MAE of 37.06 Wh and provides reliable 95% prediction intervals, while the inclusion of weather/calendar covariates proved to be mostly ineffective in reducing typical absolute error. The machine learning methods favor temporal history and calendar information, while other sensors and weather data contribute to the accuracy marginally. Chronos-Bolt has the best observed MAE of 33.58 Wh, but its advantage over the second place is within the statistical variability, and peak underestimation is severe. Thus, the choice of the best method should not be driven exclusively by the average absolute error. Considering the accuracy, simplicity, reliability, data requirements, and computational costs, SARIMA is the most compelling choice for the given forecasting task. However, Chronos should be preferred if there is an opportunity to benefit from its better average performance.

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